Phase 2 · Chunk 1 Final Report

pi vs opencode vs mini-swe-agent on 5 real SCG tickets · Claude Opus 5.5 planner, Sonnet coder · for Rohit · 2026-11-06

STATUS: COMPLETE · 18 / 18 jobs scored · 18 / 18 clean on the full regression replay

18/18	4/6 each	$2.66	$23.40
jobs scored	final passes, all harnesses	pi total cost (cheapest)	spent of the $50 budget

Evaluation progress — **18 / 18 jobs (100%)**

Budget used — **$23.40 / $50.00 (47%)**

Scoreboard (final result · cost per job)

Ticket	Type	What the fix does	pi	opencode	mini-swe-agent
SCG-140	bug	per-diem re-add uses master meal cap	**PASS · $0.15**	**PASS · $0.68**	**PASS · $0.27**
SCG-133	bug	clear advance allows zero amounts	**PASS · $0.81**	**PASS · $2.28**	**PASS · $1.58**
SCG-139	bug	payment print shows payee as requester	**FAIL · $1.35**	**FAIL · $1.78**	**FAIL · $1.26**
SCG-141	feature	petty cash payee from owner master	**PASS · $0.09**	**PASS · $0.14**	**PASS · $0.14**
SCG-137	feature	per-diem add/remove (original text)	**FAIL · $0.17**	**FAIL · $1.26**	**FAIL · $0.43**
SCG-137 v2	feature	per-diem add/remove + interface note	**PASS · $0.09**	**PASS · $0.48**	**PASS · $0.47**

Every PASS happened on turn 1. SCG-137 ran twice: v1 = original ticket text, v2 = the same ticket plus a note naming the UI labels the test checks.

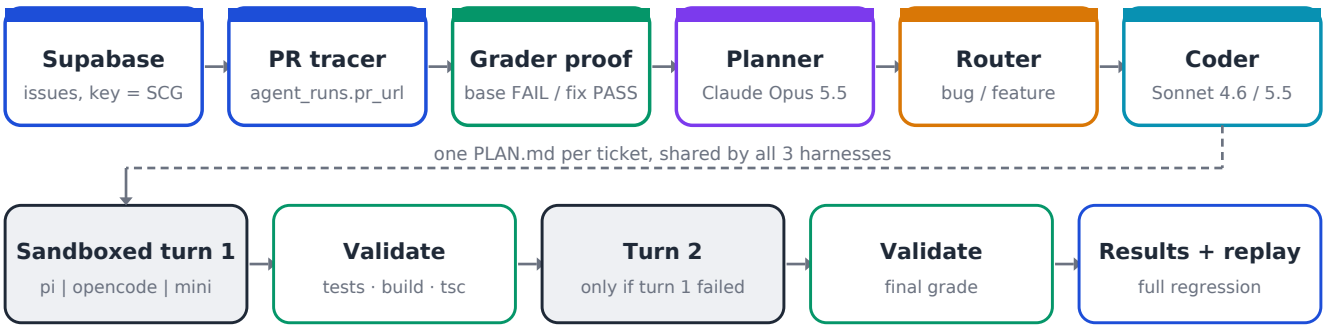
Key takeaways

- **Accuracy is tied.** Every harness scores 4/6, or 4/5 counting only the fair SCG-137 v2. The same 2 tickets failed for all 3 harnesses.
- **pi wins on cost and speed:** $2.66 and 62 min, against mini's $4.16 / 75 min and opencode's $6.61 / 102 min.
- **Failures came from the inputs, not the harness.** SCG-137 v1 lacked the label strings (v2 = 3/3 PASS). On SCG-139 the shared plan pointed at the wrong file for all 3.
- **Quality held everywhere.** The full regression replay (both suites, 1,349–1,422 tests per job) found 0 broken tests, no broken build and no new tsc errors in 18 of 18 replayed jobs.

Recommendation: use **pi** as the base for the agent-core Worker loop. It has the same accuracy at 40% of opencode's cost and 60% of its time. Invest next in **planner verification** and **ticket enrichment**: they caused every failure.

Architecture & Configuration

Pipeline



Models

Step	Model	Price / 1M (in · out)	Used for
Planner	**claude-opus-5-5**	$4 · $20	PLAN.md per ticket (5 plans, $1.43)
Coder · bug (small_task)	**claude-sonnet-4-6**	$3 · $15	SCG-140, SCG-133, SCG-139
Coder · feature (big_task)	**claude-sonnet-5-5**	$2 · $10	SCG-141, SCG-137

Harnesses

Harness	Version	Invocation	Turn 2
pi	0.84.3	pi -p --mode json, custom 5.5 models (adaptive thinking)	same session
opencode	1.19.5	opencode run --format json --auto	--continue
mini-swe-agent	2.4.6 · litellm 1.108	mini -y --exit-immediately, own $ limit	new run + failure + plan

Grading (per turn)

Check	Rule
Hidden tests	The real PR's tests: every fail-to-pass and pass-to-pass test must pass (proven on base vs. the real fix)
Regression	No test that passed on clean code may fail (touched package per turn, then a full both-suite replay of every job)
Build · typecheck	`pnpm run build` must not newly break. `tsc` must add no new errors (path-normalized vs. baseline)
Real change	At least 1 file modified, so a no-op can never pass

Performance · 18 jobs

Harness	Final pass	Fair (v2) pass	Turn-1 pass	Total cost	$ / pass	Wall time	Cache read	Output
pi	4/6	4/5	4/6	$2.66	$0.67	62 min	4.0 M	52 k
mini-swe-agent	4/6	4/5	4/6	$4.16	$1.04	75 min	7.7 M	70 k
opencode	4/6	4/5	4/6	$6.61	$1.65	102 min	13.1 M	83 k

Total cost (USD)



Total wall time (min)



Per job (turns · cost · time)

Ticket	Coder model	pi	opencode	mini-swe-agent
SCG-140	sonnet-4-6	**PASS · 1t · $0.15 · 7m**	**PASS · 1t · $0.68 · 8m**	**PASS · 1t · $0.27 · 7m**
SCG-133	sonnet-4-6	**PASS · 1t · $0.81 · 9m**	**PASS · 1t · $2.28 · 15m**	**PASS · 1t · $1.58 · 15m**
SCG-139	sonnet-4-6	**FAIL · 2t · $1.35 · 20m**	**FAIL · 2t · $1.78 · 21m**	**FAIL · 2t · $1.26 · 19m**
SCG-141	sonnet-5-5	**PASS · 1t · $0.09 · 6m**	**PASS · 1t · $0.14 · 6m**	**PASS · 1t · $0.14 · 7m**
SCG-137	sonnet-5-5	**FAIL · 2t · $0.17 · 12m**	**FAIL · 2t · $1.26 · 40m**	**FAIL · 2t · $0.43 · 15m**
SCG-137 v2	sonnet-5-5	**PASS · 1t · $0.09 · 8m**	**PASS · 1t · $0.48 · 12m**	**PASS · 1t · $0.47 · 13m**

What the numbers say

- **The same model gives a 2.5× cost gap.** opencode read 13.1 M cached tokens against pi's 4.0 M, for the same tickets and the same plans.
- **Turn 2 rescued 0 of 6 attempts.** Every pass came on turn 1. On SCG-139, the assertion feedback moved pi to the right file (`print-data.ts`), but its logic was still wrong.
- **Ticket clarity decides the outcome.** The same feature went from 0/3 (v1) to 3/3 on turn 1 (v2) once the 2 UI label strings were in the ticket.
- **The planner is a single point of failure.** The SCG-139 plan named the web form, not the printed form, and every harness followed it.

Full Regression Replay & Shared Build Directory

Each job's final patch was re-applied to its base commit, and both full suites (web + app) were run with the build and the typecheck. $0, no model calls. The hidden PR test files are excluded.

Job	Web tests	App tests	Broken	Build	New tsc	Replay
SCG-140-pi	732 / 732	651 / 651	0	OK	+0	6 m 57 s
SCG-140-opencode	732 / 732	651 / 651	0	OK	+0	6 m 58 s
SCG-140-mini	732 / 732	651 / 651	0	OK	+0	6 m 59 s
SCG-133-pi	715 / 715	646 / 646	0	OK	+0	8 m 59 s
SCG-133-opencode	715 / 715	647 / 647	0	OK	+0	9 m 08 s
SCG-133-mini	716 / 716	647 / 647	0	OK	+0	7 m 07 s
SCG-139-pi	729 / 729	620 / 620	0	OK	+0	5 m 39 s
SCG-139-opencode	730 / 730	620 / 620	0	OK	+0	6 m 34 s
SCG-139-mini	730 / 730	620 / 620	0	OK	+0	6 m 44 s
SCG-141-pi	729 / 729	693 / 693	0	OK	+0	7 m 03 s
SCG-141-opencode	729 / 729	693 / 693	0	OK	+0	7 m 07 s
SCG-141-mini	729 / 729	693 / 693	0	OK	+0	7 m 17 s
SCG-137-pi	724 / 724	651 / 651	0	OK	+0	6 m 07 s
SCG-137-opencode	724 / 724	652 / 652	0	OK	+0	5 m 42 s
SCG-137-mini	724 / 724	651 / 651	0	OK	+0	6 m 13 s
SCG-137-pi v2	724 / 724	651 / 651	0	OK	+0	8 m 29 s
SCG-137-opencode v2	724 / 724	651 / 651	0	OK	+0	8 m 27 s
SCG-137-mini v2	724 / 724	651 / 651	0	OK	+0	6 m 59 s

Shared build directory (measured on the sandboxed jobs)

Metric	Per job (average)	Across 7 sandboxed jobs
Setup time	5.9 s	41 s
node_modules shared by hard link (no new disk)	420 MB · 41 k files	2.9 GB not copied
New disk (source tree + 1-commit repo)	74 MB	0.51 GB
node_modules dirs linked	12	web · app · appforge workspace

Method: `git archive` of the base commit + `cp -al` of each node_modules from the job's prepared worktree (pnpm store → worktree → /tmp). Build output goes outside the worktree. The earlier jobs used the same hard-linked worktrees without measuring.

Incidents & the Rootless Sandbox

Isolation incidents (all caught, all invalidated runs re-done)

When	What the logs showed	Action
11-05 22:14	Non-isolated worktrees. On SCG-139, pi read and copied opencode's test files, opencode ran `find` over the research dir, and mini worked in a Phase 1 worktree	**3 runs invalidated**
11-05 22:50	Stopped. Every log scanned: no .env, token, key or ~/.ssh access. The Phase 1 worktree was restored	**no secret exposure**
11-06 09:59	Sandboxed run: mini set `REPO=/tmp/eval-job-…opencode…` and worked in a finished sibling job's /tmp dir	**1 run invalidated**
11-06 10:13	Sibling /tmp dirs locked + guard on other jobs' paths. All later runs: no path outside their own dir	**fixed · verified**

The 5-layer rootless sandbox (no sudo; user namespaces blocked by AppArmor)

1 — Isolated /tmp workspace
git archive of the base + hard-linked node_modules + a fresh 1-commit git repo: no history, no remotes, no link back.

2 — Command guards (PATH)
chmod blocked. grep, rg and find refuse '..'. All refuse /home/mukesh paths and any other job's /tmp/eval-job-* path.

3 — Directory lock
Research dir chmod 0000 while a harness runs, restored to 775 afterwards (also on crash or exit). Private HOME and XDG dirs.

4 — Sibling job lock
Every other /tmp/eval-job-* dir (finished ones too) chmod 0000 during the turn, then restored.

5 — Prompt sanitization
Turn-2 feedback lists failing checks as assertions. No mention of "hidden" or "acceptance" tests.

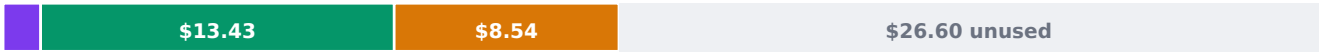
Verification (free tests, research dir locked)

Escape attempt	Result	Real work inside the sandbox	Result
cat .env · ls the research dir · Phase 1 worktree	**Permission denied**	vitest file	**29/29 pass**
ls / cat / grep another job's /tmp dir	**blocked + denied**	vite build	**built**
find .. · chmod 755	**blocked**	node:test (tsx)	**3/3 pass**
git log / branches	**1 commit, main**	pi · opencode · mini	**reach Anthropic**

Limit: same-uid hardening stops wandering agents, not a determined attacker. Containers (Docker or bubblewrap) need sudo and remain the long-term fix. With the global lock, jobs run one at a time.

Budget, Lessons & Next Steps

Where the $23.40 went



Planner (5 plans) · 18 scored jobs · Invalidated, aborted and rerun turns

Bar scale = the full $50 budget

Engineering issues found & fixed

Issue	Fix	Status
JUnit parse broke on an unescaped '>' in test names	Quote-aware parser + re-parse of saved reports	fixed
tsc messages embed each worktree's absolute path	Normalize paths before the baseline comparison	fixed
mini first-run wizard; missing mcp package	MSWEA_CONFIGURED + mcp in the venv	fixed
Agents could read sibling worktrees, then sibling /tmp dirs	5-layer rootless sandbox	fixed · verified
Anthropic custom spend limit reached mid-run	Limit raised; resumed with kept records and prior spend	resolved
Disk at 96–98% blocked Run B	Removed finished /tmp job dirs (patches + logs kept)	resolved
Ticket text missing the interface strings the test checks (SCG-137)	v2 interface note; 3/3 PASS	validated

Recommendations for agent-core

- **Worker harness: pi.** It has equal accuracy, the lowest cost per pass and the shortest runs. Keep opencode and mini as benchmarks only.
- **Verify the plan before coding.** Add a cheap check that the files named in PLAN.md match the ticket's surface (for example, print vs. screen).
- **Enrich tickets at triage.** Attach interface strings, routes and field names the QA tests use, as Ridge's triage metadata already partly does.
- **Isolation by default.** Run every agent in a container or the rootless sandbox. Never run in a linked worktree under the shared research dir.

Next steps (Chunk 2)

	Step	Why
☐	20+ tickets, 2 runs per job	3 harnesses tie on 5 tickets: more samples are needed to separate accuracy
☐	`vitest --related` per turn + nightly full replay	Cuts validation from ~6 min to about 1 min per turn
☐	Container sandbox (needs sudo once)	Real isolation and parallel jobs again
☐	Planner-verification step	Removes the single point of failure seen on SCG-139